PORTFOLIO 1: DATABASE DESIGN AND MODELLING

NaijaFresh Online Store – A Simulated Nigerian E-commerce Grocery Platform

1. Introduction and Case Study Description

NaijaFresh is a simulated Nigerian online grocery store designed to provide customers with a convenient platform for purchasing groceries online.

The database was developed to manage the store's major operations, including customer registration, product categorisation, inventory records, customer orders, order items, payments, deliveries and product reviews.

The purpose of the database is to organise business information, establish relationships between related records, minimise unnecessary data duplication and support efficient data retrieval.

All customer and transaction information used in this project is fictional and created for academic purposes.

2. Database Objectives

The objectives of the NaijaFresh database are to:

Store and manage customer information.

Organise grocery products into categories.

Maintain product prices and stock quantities.

Record customer orders and the products purchased.

Maintain payment information and transaction references.

Track delivery addresses and delivery statuses.

Store customer ratings and product reviews.

Maintain data integrity through keys and constraints.

3. Database Tables and Their Functions

The database consists of eight tables.

4. Primary Keys and Foreign Keys

A primary key uniquely identifies each record in a database table. A foreign key connects a table to another table by referencing its primary key.

The following keys were implemented:

The primary keys use AUTO_INCREMENT to generate unique identifiers automatically.

5. Database Relationships

The database contains the following relationships:

Customers and Orders – One-to-Many

One customer can place multiple orders, while each order belongs to one customer.

Categories and Products – One-to-Many

One category can contain multiple products. Each product references a category.

Orders and Order Items – One-to-Many

One order can contain multiple products. The order_items table records each product separately.

Products and Order Items – One-to-Many

A product can appear in multiple order items across different customer orders.

Customers and Reviews – One-to-Many

A customer can submit multiple product reviews.

Products and Reviews – One-to-Many

A product can receive multiple reviews from customers.

Orders and Payments – One-to-Many

The database permits multiple payment records for an order because the order_id column in the payments table is not unique.

Orders and Deliveries – One-to-One

Each order can have only one delivery record because order_id in the deliveries table has a UNIQUE constraint.

6. Database Constraints

Several constraints were implemented to maintain data integrity.

PRIMARY KEY: Ensures each record has a unique identifier.

FOREIGN KEY: Maintains valid relationships between related tables.

NOT NULL: Prevents required fields from being left empty.

UNIQUE: Prevents duplicate values in selected columns, including customer email addresses and payment transaction references.

DEFAULT: Supplies automatic values where appropriate, including timestamps and default statuses.

CHECK: Restricts product review ratings to values between 1 and 5.

For example, the reviews table uses:

CHECK (rating BETWEEN 1 AND 5)

This prevents invalid ratings outside the permitted range.

7. Database Normalisation

Normalisation is the process of organising database tables to reduce unnecessary duplication and improve data integrity.

First Normal Form (1NF)

The database uses individual columns containing single values rather than multiple values in one field.

For example, products purchased in one order are stored as separate records in the order_items table instead of being combined into a single products column in the orders table.

Second Normal Form (2NF)

The tables use individual primary key columns to identify records. The design avoids partial dependencies on composite primary keys.

Third Normal Form (3NF)

The database separates customer, product, category, order and payment information into different tables. This reduces unnecessary duplication and makes information easier to maintain.

The orders table stores total_amount as a transaction summary. Since this amount can also be calculated from the associated order_items, the application must ensure that the stored total remains consistent with the item quantities and prices.

8. Implementation and Testing

The database was implemented using MySQL Server and MySQL Workbench.

The following practical activities were completed:

Created the naijafresh_db database.

Created eight related database tables.

Implemented primary keys and foreign keys.

Inserted five fictional customer records.

Inserted five product categories.

Inserted ten grocery products.

Created three sample customer orders.

Inserted eight order item records.

Inserted three payment records.

Inserted three delivery records.

Inserted five customer reviews.

Executed SELECT queries to retrieve records.

Executed a JOIN query to retrieve information from four related tables.

Reverse-engineered the database in MySQL Workbench to generate an EER diagram.

The JOIN query successfully combined information from the orders, customers, order_items and products tables.

9. Conclusion

The NaijaFresh database demonstrates how relational database design can support the operations of an online grocery store.

The implementation established eight related tables with primary keys, foreign keys and integrity constraints. Sample records were inserted and tested using SQL queries.

The project provided practical experience in database creation, relational modelling, data insertion, querying, relationships and normalisation.

10. Tools Used

MySQL Server

MySQL Workbench

SQL

GitHub (for version control and portfolio submission)

11. AI Assistance Declaration

ChatGPT was used as a learning and support tool to explain database concepts, guide SQL implementation, assist with troubleshooting and support documentation.

The SQL statements were executed in MySQL Workbench, and the resulting database structures and query outputs were checked by the student. The student is responsible for understanding and explaining the implementation during technical verification.

Table	Purpose
customers	Stores customer information, including names, email addresses, phone numbers and addresses.
categories	Stores grocery product categories.
products	Stores product names, descriptions, prices and stock quantities.
orders	Records customer purchases and order information.
order_items	Stores individual products included in each order.
payments	Records payment methods, amounts, statuses and transaction references.
deliveries	Stores delivery addresses and delivery progress.

Table	Purpose
reviews	Stores customer product ratings and feedback.

Table	Primary Key	Foreign Key(s)
customers	customer_id	None
categories	category_id	None
products	product_id	category_id
orders	order_id	customer_id
order_items	order_item_id	order_id, product_id
payments	payment_id	order_id
deliveries	delivery_id	order_id
reviews	review_id	customer_id, product_id